

Lightning Potential Index (LPI)

Calculation and display in the BC HRDPS Fire Weather plot

EXPERIMENTAL INGREDIENTS INDEX

WHAT IT MEANS

A 0-100 relative measure of overlap among **instability**, **mixed-phase moisture**, and evidence that convection is developing or occurring. The same calculation is used in the HRDPS 2.5-km BC and HRDPS-West 1-km regional Fire Weather plots. It is **not** a lightning probability or flash-density forecast.

CORE CALCULATION

$$\text{LPI} = 100 \times I \times M \times \text{sqrt}(T)$$

I = instability | M = mixed-phase moisture | T = realization / trigger

For every threshold below: $R(x; a, b) = \text{clip}[(x-a)/(b-a), 0, 1]$. R is zero at a, one at b, and linear between them. After multiplication, LPI is set to zero if $M < 0.08$, or if both the LI factor < 0.05 and CAPE factor < 0.08 ; the hourly field is smoothed about 1.2 km and clipped to 0-100.

1 INSTABILITY (I)

$$I = R(1-LI; 0, 6) \times [0.80 + 0.20 R(\text{CAPE}; 75, 800)]$$

MU-LI is primary: its factor starts below +1 C and reaches 1 at -5 C. Examples: LI 0 = 0.17; -2 = 0.50; -4 = 0.83.

Most-unstable CAPE ramps from 75 to 800 J/kg, but can strengthen I by only 0-20%. CAPE cannot create LPI without LI support.

2 MIXED-PHASE (M)

RHcharge is the mean RH of above-ground model layers from 0 to -20 C, weighted by each layer's pressure thickness and most strongly near -15 C. It is a proxy for a saturated mixed-phase cloud environment where ice/liquid interactions can separate charge.

$$C = \text{sqrt}[R(\text{RHcharge}; 45, 80) \times R(\text{depth}; 35, 150 \text{ hPa})]$$

Depth is the sum of those same temperature-weighted pressure thicknesses. More depth means more mixed-phase cloud volume and opportunity for charging. The geometric mean requires both a humid and a deep layer.

$$M = C \times [0.85 + 0.15 R(\text{RHmid}; 35, 75)]$$

RHmid from -5 to -30 C can add only 0-15%.

3 REALIZATION (T)

Question: is convection actually developing or occurring, rather than merely possible?

Three independent clues are scaled from 0 to 1: trailing 3-h precipitation (0.05-1.5 mm), current precipitation rate (0.02-0.8 mm/h), and resolved ascent at 500/700 hPa (geometric w 0.005-0.050 m/s) occurring with a charging layer.

$$T = \text{max}(\text{P3h factor}, \text{rate factor}, \text{w factor} \times C)$$

Taking the maximum lets model precipitation confirm an existing storm, while ascent plus C can identify development before appreciable rain. sqrt(T) keeps an early/weak clue from reducing LPI too severely.

DRY-LIGHTNING POTENTIAL

$$\text{RHdry} = \min(\text{RH2m}, \text{RHsubcloud})$$

$$D = R(55 - \text{RHdry}; 0, 25)$$

$$\text{Dry3h} = \text{max}\{1[\text{LPIh} \geq 20] \times \text{LPIh} \times \text{Dh}\} \times [1 - R(\text{P3h}; 0.25, 2.5)]$$

RHsubcloud is the mean of 2-m RH and valid above-ground RH at 850/800/750/700 hPa. Dryness begins below 55% and is full at 30% or less. Reduction of dry lightning potential by rainfall begins above 0.25 mm per 3 h and reaches zero at 2.5 mm.

A dry-lightning asterisk is plotted where Dry3h ≥ 15 . There is no second omega gate: LPI already contains the convective trigger.

HOW LPI APPEARS IN THE FIRE WEATHER PLOT

- Look at the **right panel**. Fire danger is the background fill. LPI is shown only as **purple-to-magenta contours at 20, 40, 60 and 80**; lines become thicker and more magenta with increasing LPI. The contour field receives an additional 5-km smoothing for readability.
- Asterisks appear where Dry3h ≥ 15 . They are **grey where LPI is below 60 and black at LPI 60 or greater**. LPI 20 is the internal dry-lightning eligibility gate, not a universal thunderstorm threshold.
- Blue dots show the same ending 3-h rainfall used in the dry-lightning calculation: pale cyan at 2.5-10 mm and teal at 10 mm or more. This rainfall context is separate from the LPI contours.
- For F003, F006, etc., LPI and dry-lightning potential are the **maximum of the three hourly calculations** ending at valid time; F000 is instantaneous. The left panel shows RH and gust, not LPI. Use the index with radar/satellite, forcing and soundings; calibration is ongoing.